

Teacher

1.2 Classwork: Arithmetic Sequences

1. Consider the sequence

7, 11, 15, 19, ...

- (a) Write down the value of u_1 , the first term.
- (b) State whether the sequence is arithmetic. Give a reason for your answer.
- (c) Write down the value of the common difference d .
- (d) Write down the next two terms of the sequence.
- (e) Find an expression for u_n , the n^{th} term.
- (f) Find u_{50} .

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(a) 7

(b) Yes: $11 - 7 = 15 - 11 = 19 - 15 = 4$
(same) common difference

(c) $d = 4$

(d) 23, 27

(e) $u_n = 7 + 4(n-1)$

(f) $u_{50} = 7 + 4(50-1)$
 $= 203$

2. An arithmetic sequence has first term $u_1 = 40$ and common difference $d = -3$.
- (a) Write down the first four terms of the sequence.
 - (b) Find u_{20} .
 - (c) One term of the sequence is equal to -50 . Find the value of n for that term. Show your algebraic working.

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(a) $40, 37, 34, 31$

(b) $u_{20} = 40 + (-3)(20-1)$
 $= -17$

(c) $u_n = 40 + (-3)(n-1) = -50$
 $-3(n-1) = -90$
 $n-1 = 30$
 $n = 31$

check $u_{31} = 40 + (-3)(31-1)$
 $= -50 \quad \checkmark$